

1 Inputs

Complex problems
and questions

Evaluate the integral:

$$I = \int_0^{\pi} x \sin(x) dx$$

Prove or disprove:

$\forall n \in \mathbb{N}, n^2 + n$ is even

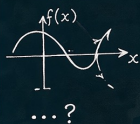
Logic puzzle:

A says: "B is lying."
B says: "C is lying."
C says: "A is telling the truth."
Exactly one of them is telling the truth. Who is it?

Optimization:

$$\max_x f(x) = x^3 - 3x + 2$$

subject to $x \in [-2, 2]$



Problem types

- ✓ Mathematical reasoning
- ✓ Logical puzzles
- ✓ Proofs and theorems
- ✓ Optimization problems
- ✓

2 Thinking LLM Core

Deliberate, explore and self-correct



Tree of
thoughts



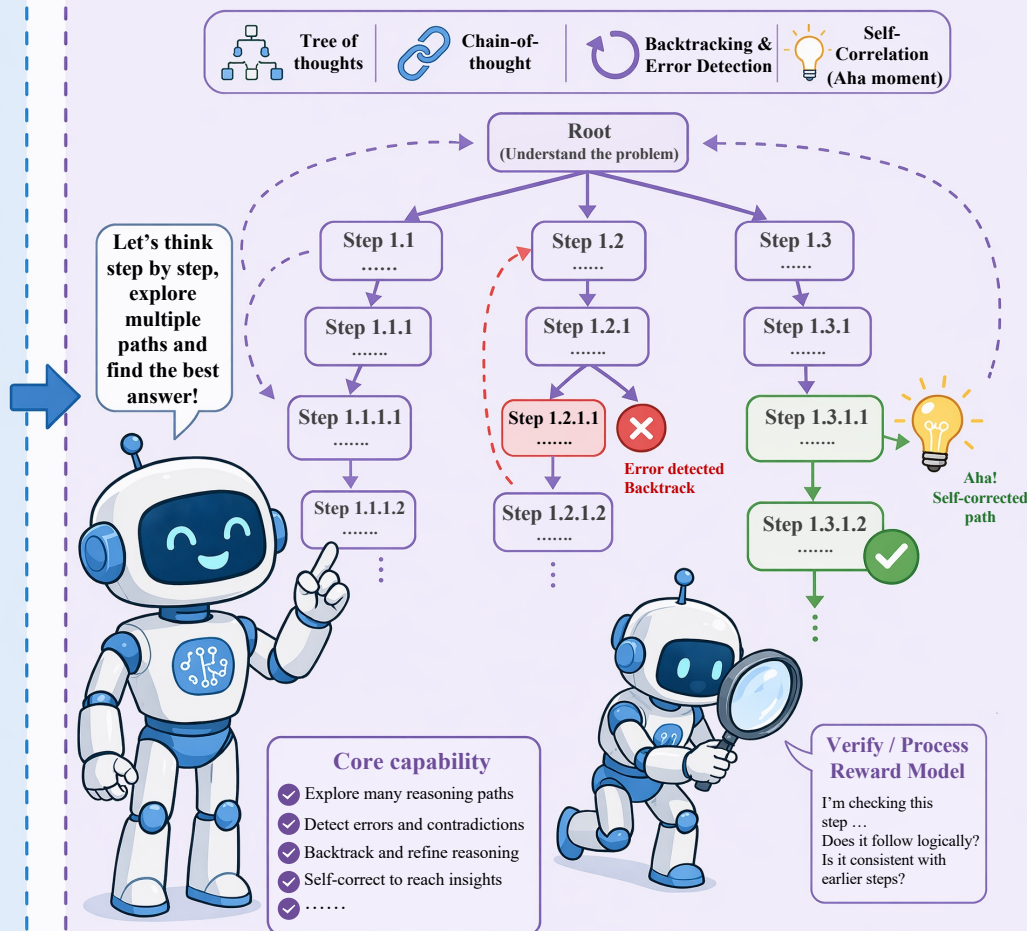
Chain-of-
thought



Backtracking &
Error Detection



Self-
Correlation
(Aha moment)



Let's think
step by step,
explore
multiple
paths and
find the best
answer!

Core capability

- ✓ Explore many reasoning paths
- ✓ Detect errors and contradictions
- ✓ Backtrack and refine reasoning
- ✓ Self-correct to reach insights
- ✓

Verify / Process Reward Model

I'm checking this
step ...
Does it follow logically?
Is it consistent with
earlier steps?

3 Outputs

Structured reasoning and
find answer

Reasoning Tree (Chain-of-thought)

- 1 Understand the problem and define goals.
- 2 Recall relevant formulas and theorems.
- 3 Consider possible approaches.
- 4 Explore Approach A ... (discarded)
- 5 Explore Approach B ...
- 6 Derive key intermediate result.
- 7 Check for errors and edge cases.
- 8 Refine the reasoning.
- 9 Arrive at the correct solution.
- ...

Final Answer



$$I = \pi$$

That make sense!
Clear, verifiable
and correct

